ModelIt!

Student Activity Pack

The Ultra-Processed Food Problem

Modeling Food Chemistry, Metabolic Pathways, and Nutritional Systems

Grade 12L1 | NGSS HS-LS1-7 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What did you eat for your last meal? Try to list every ingredient you can think of.

2 What do you think the difference is between 'processed' and 'ultra-processed' food?

3 Why do you think some foods are very easy to overeat while others fill you up quickly?

4 Do you think the cost of food affects what people eat? How?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Processing Level | Sugar and Additive Load | Satiety Response | Metabolic Health | Caloric Overconsumption

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Whole Foods

Processing: Minimal | All meals from whole ingredients

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Ultra-Processed

Processing: Maximum | 80% ultra-processed calories

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Diet Shift

Processing: Transition from 80% UPF to 50% whole foods over 4 weeks

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Gut Microbiome Diversity — The variety of bacterial species in the digestive tract — decreases with ultra-processed food consumption and correlates with metabolic health, immune function, and even mental health

Food Marketing Exposure — The amount and type of food advertising a person encounters — children and teens see an average of 13 food ads per day, overwhelmingly for ultra-processed products

Socioeconomic Access — The availability and affordability of whole foods in a person's neighborhood — food deserts with limited fresh food access correlate strongly with higher ultra-processed food consumption

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why do foods engineered to be irresistible in your mouth end up being catastrophic for your body — and how did we build a food system where the cheapest calories are the most dangerous? Use evidence from your simulation to support your answers.

1. **How does your model explain why calorie counting alone doesn't capture the health impact of different diets?**

2. **Why is it misleading to blame obesity on 'lack of willpower' based on what you've learned about satiety signaling?**

3. **What does the model suggest about the responsibility of food companies versus individual choice?**

4. **How might understanding food processing change the way you read nutrition labels?**

5. **What are the ethical implications of engineering foods to bypass biological satiety signals?**

8 STEM Challenge: Design a School Lunch That Beats Ultra-Processing

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YOUR ENGINEERING MISSION

Design a school lunch program that provides meals as satisfying and convenient as ultra-processed options but using minimally processed, whole-food ingredients — within the current per-meal budget.

THE SCENARIO

Your school district spends $1.50 per student per meal on ingredients, and 75% of current menu items are ultra-processed. The district wants to reduce ultra-processing to 25% while maintaining cost, student acceptance, and nutrition standards. Your team must design a week of lunch menus with a cost analysis.

Engineering Design Thinking

1. WHAT SPECIFIC MEALS COMPRISE YOUR WEEKLY MENU AND WHAT IS THE NOVA PROCESSING CLASSIFICATION OF EACH COMPONENT?

2. HOW DOES YOUR TOTAL PER-STUDENT COST COMPARE TO THE CURRENT BUDGET, AND WHAT TRADE-OFFS DID YOU MAKE?

3. WHAT EVIDENCE FROM YOUR MODEL SUPPORTS THE CLAIM THAT YOUR MENU WILL IMPROVE STUDENT HEALTH OUTCOMES?

4. HOW WOULD YOU MEASURE STUDENT ACCEPTANCE AND HEALTH IMPACTS OF YOUR NEW MENU?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Food scientists study the chemistry of food processing and develop healthier alternatives, earning $60,000-$100,000/year. Registered dietitians design nutrition programs for schools, hospitals, and communities, earning $55,000-$85,000/year. Public health nutritionists research diet-disease relationships at $65,000-$110,000/year.